

Advanced Wireless Communication CT548 (3-0-2-4)

Instructor: Dr. Manish Kumar

Course Placement: It is a Technical elective course for MTech first year (second semester) students and BTech third year (sixth semester) students.

Course Pre-requisite: Fundamental understanding of courses like Probability and Statistics, Signals and Systems, and Digital Communication.

Course format: It would be 150 minutes' lecture and 2-hour lab session (coding) every week.

Course content: The course primarily starts with the evolution of wireless technologies, followed by wireless channel modeling and representation. Later, important concepts like MIMO, OFDM which emerged as key technologies for 4G and 5G wireless communications are discussed. Important aspect of channel estimation for wireless communication systems will be discussed in detail. Subsequently, the Massive MIMO and milli-meter wave systems will be discussed for next-generation wireless communication systems. The prime motive of the course is to provide an understanding and background in the domain of wireless communication which could pave the way forward to academic research in various areas of wireless communication. Certain modules of the course could be helpful and serve as an important pre-requisite in the relevant companies working in the domain of wireless communication.

Few Text Books and Reference Books:

1. "Wireless Communications", Andrea Goldsmith, Cambridge University Press.
2. "Fundamentals of Wireless Communication", David Tse & Pramod Vishwanath, Cambridge
3. "MIMO Wireless Communications", Ezio Biglieri et. al., Cambridge University Press.
4. "Wireless Communication Principles and Practice" T Rappaport

Assessment Method: Continuous evaluation will be carried out throughout the semester with Mid-Semester Examination, End-Semester Examination and some weightage to the assignment. There might be a very small change in this assessment method but in that case it would be duly notified well in advance.

Laboratory Component: Students will be given weekly problems which need to be simulated in MATLAB. Focus will be on the different components of the system under study. Tentatively, a report containing the results need to be prepared for evaluation. The students can either use the institute MATLAB or use a personal license. The idea is to enhance coding expertise and understanding of wireless and digital communication system and their modeling.

Course Objectives:

- After the successful completion of the course it would lay forward the foundation towards working upon further academic research problems and assignment in the broad domain of wireless communication and aligned research areas.
- The candidates will gain important theoretical concepts whose knowledge is mandatory for fresh graduates working in the relevant wireless communication and associated industries. The MATLAB programing learnt during the practical sessions would be of prime help in correlating and better understanding of the fundamentals of wireless communications.
- Problem Analysis and Design of Systems: The candidate would be able to utilize the concepts learnt in the course in various principles of wireless communications like algorithm design, performance

evaluation and coding assignments in Industry which requires theoretical understanding of the overall wireless transceiver architecture.

Course Outcomes:

- To enable the students to develop a strong background in wireless communication, understand the overall wireless system model and use the tools learnt in order to solve the complex algorithmic design problem.
- Apart from further academic research in wireless communication where the course would be a good background to proceed further, the knowledge would of significant help if the graduates look forward to work in the relevant wireless communication industries be it 5G system engineering team, algorithm, Modem system design and several other relevant teams as well.
- The objective is to enables the graduates to take individual responsibility and work as a part of a team towards the fulfillment of both individual and organizational goals in such assignments where the subject knowledge can act as a major advantage.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X						X			X

Lecture Schedule:

1. **WSSUS Channel Model, Diversity & Capacity Analysis of Fading Channels [10 hours]**

- 1.1. WSSUS Model: Doppler, Temporal Characteristics of Fading Channel
- 1.2. Classification of Fading Channels: Coherence time and Bandwidth
- 1.3. Effect of Mobility and Doppler Spectrum
- 1.4. Introduction to Principles of Diversity
- 1.5. Antenna Diversity and Maximal Ratio Combining
- 1.6. BER Analysis of Multi Antenna Systems
- 1.7. Deep Fade in Multi Antenna System, Diversity Order
- 1.8. Capacity of Fading Channels

2. **MIMO Systems: [8 hours]**

- 2.1. Introduction: MIMO System Model
- 2.2. MIMO Receivers and Performance Analysis
- 2.3. Beamforming and Space Time Block Codes
- 2.4. Singular Value Decomposition (SVD) in MIMO Systems
- 2.5. SVD Based MIMO Transmission
- 2.6. Capacity of MIMO Systems

3. **OFDM Systems: [6 hours]**

- 4.1 Multicarrier Modulation and Transmission
- 4.2 Introduction to OFDM
- 4.3 Cyclic Prefix and Signal Model
- 4.4 OFDM Transmitter and Receiver Models
- 4.5 Performance Evaluation of OFDM Systems

4. **Channel Estimation Approaches in Wireless Communication Systems: [8 hours]**
 - 4.1. Maximum Likelihood (ML) and Bayesian Channel Estimation for SISO systems
 - 4.2. ML and Bayesian Channel Estimation for MISO systems
 - 4.3. ML and Bayesian Channel Estimation for MIMO systems

5. **Introduction to Massive MIMO and mmWave Technologies: [10 hours]**
 - 5.1. Massive MIMO System Architecture-I
 - 5.2. Massive MIMO System Architecture-II
 - 5.3. mmWave System Architecture-I
 - 5.4. mmWave System Architecture-II
 - 5.5. Discussions on Relevant 5G Related 3GPP Technical Specification & usage in Industry